

Riley Beenders

MECHANICAL DESIGN | DESIGN COORDINATION | ENGINEERING LEADERSHIP

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PROFESSIONAL SUMMARY

Mechanical and electromechanical engineering leader with 5+ years of professional engineering experience and 10+ years of multidisciplinary project work. Owns complex mechanical scopes from concept design and drawings through vendor coordination, procurement, fabrication, assembly, and field support. Brings hands-on regulatory compliance work, investigation of unfamiliar mechanical and electrical problems, engineering hiring, and consensus-building across mechanical, electrical, software, and manufacturing disciplines.

CORE EXPERTISE

Mechanical Design: SolidWorks; Autodesk Inventor; Siemens NX; CATIA; Fusion 360; GD&T; design for manufacturing and assembly (DFM/DFA); fixture and tooling design; thermal and acoustic enclosure design

Systems & Analysis: Motion and drive systems; gantry and machine design; electrical layout and schematic design; MATLAB, Simulink, and Simscape; R&D prototyping and validation; data logging and process analysis

Project Delivery: Design coordination with vendors and fabricators; procurement, sourcing, and logistics; engineering documentation and BOM control; vendor communication and cost negotiation; field troubleshooting and support

Compliance & Leadership: Product safety and regulatory compliance (2006/42/EC machinery directive, EU and U.S. standards); risk analysis; technical writing; Lean Six Sigma; hiring and technical interviews; worldwide project collaboration

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Led the design and development of ICARUS-Lite, owning its mechanical, electrical, and software design, including custom AC heaters and an IEEE-style electrical design adaptable to 120/240 VAC.
- Addressed CE Machinery Directive requirements on ICARUS-Lite and completed product changes supporting conformity with 2006/42/EC, while reducing assembly time by 80% compared to previous designs.
- Re-engineered POSEIDON and ICARUS with a standardized linear-motor architecture and swappable mounts, replacing servo-driven ballscrews, demonstrating 4-micron precision at 4 m/s with a 30 kg moving mass, and reducing support complexity.
- Co-developed a custom hotend with an external manufacturing partner, improving reliability and reducing cost for the company's materials.
- Built an automated LabVIEW QC system that negotiates Modbus communication with three tri-axial laser sensors over a dedicated private VLAN, collecting nine data points (three per laser) every 17.5 ms and charting live tolerances.
- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.
- Owned ICARUS-Lite through commercialization with plug-and-play setup, contributing to increased revenue.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.
- Provide cross-functional technical support for the product line, from troubleshooting to after-sales service.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer

January 2023 - January 2025

- Led the design and launch of Gen3 POSEIDON, combining a 1 m³ build volume, filament-and-pellet dual extrusion, braked NEMA 34 Z drives that prevent uncontrolled motion during power loss, industrial-servo XY motion, and a custom gantry.
- Researched high-speed linear motors that moved a 30 kg payload at 3 m/s with 5-micron accuracy and conducted early 6-axis robotic non-planar printing research.

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while retaining hands-on responsibility for mechanical, electrical, firmware, and software systems.

Lead Electro-Mechanical Engineer

June 2022 - January 2023

- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.
- Developed independently testable electronics enclosures that could be installed in minutes and coordinated custom harnesses, panel designs, and hardware improvements with vendors.
- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging while supporting larger, modular machines.

Industrial Engineer

April 2021 - June 2022

- Designed THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry system for a large-scale statue client, and managed CAD, BOM, sourcing, fabrication, assembly, firmware tuning, support, and training.
- Developed next-generation ICARUS and POSEIDON models with higher-performance components and expanded motion safety through NEMA 34 motors, planetary gearboxes, and power-failure brakes.
- Designed and released Kratos, a machine built for high-strength above-knee and below-knee prosthetic sockets.

Engineering Technician

May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized process parameters while documenting design limitations and opportunities for product improvement.

SELECTED PROJECTS

Linear Motor Platform Standardization | Proteor | Converted both flagship platforms from servo-driven ballscrews to a shared linear-motor architecture with swappable mounts, and validated 4-micron precision at 4 m/s with a 30 kg moving mass on the machine platforms.

ICARUS-Lite - Product Development | 2025 - 2026 | Compact industrial system for prosthetic and orthotic production: bolt-less chamfered shell, 120/240 VAC input, plug-and-play setup, 80% faster assembly than previous designs, and 2006/42/EC machinery directive changes.

Gen3 POSEIDON - Production-Focused Machine Redesign | Filament Innovations | Led mechanical, electrical, firmware, and system-level design of the flagship platform's third generation: 1 m³ build volume, hybrid filament-and-pellet extrusion, industrial-servo XY motion, and braked NEMA 34 Z motors that prevent uncontrolled motion during power loss.

THE KRAKEN - Custom Hybrid Gantry System | Filament Innovations | Delivered a one-off 1 x 2 x 1.5 m pellet-and-filament gantry machine for a customer's large-scale statue project, from machine CAD and a scalable bill of materials through sourcing, fabrication coordination, assembly, firmware tuning, and customer training.

Modular CAN-Bus Controls Architecture | Filament Innovations | Self-contained, independently testable electronics enclosures connected over CAN bus, separating electrical build work from final machine integration and cutting installation of tested electronics to minutes.

ADDITIONAL RELEVANT EXPERIENCE

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and operate a small online store for printer heat inserts and accessories; produced plastic face shields during the COVID-19 pandemic.

Blue Mountain Ski Resort - Ski Lift Operator | November 2019 - January 2021 | Operated passenger lifts, performed motor and control safety checks, and supported the maintenance team during lift breakdowns while assisting thousands of guests.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Certifications: MathWorks MATLAB, Simulink, Simscape, and Machine Learning Onramps (2026); Lean Six Sigma Yellow Belt and White Belt, Educate 360 (2025); Penn State Certificate in Engineering Design and Tools (2023); Lafayette College Mechanical Engineering Certificate (2019)